Repository

SnapMeal AI

Demo

Introduction

Ever ordered food online and thought, “This looks good… but what’s actually in it?” SnapMealAI turns that uncertainty into clarity. just snap a photo of any dish or menu item and instantly get personalized insights on ingredients, calories, allergens, and healthier alternatives. Whether you're browsing a delivery app or scanning a restaurant menu, SnapMealAI helps you make smarter food choices with confidence.

Stakeholders

- **Consumers** who want instant nutritional and allergen insights for meals and menus.
- **Restaurants** seeking to provide transparent nutrition information to customers.
- **Food delivery services** integrating AI-based meal analysis for added user value.
- **Healthcare & wellness partners** who want to monitor or recommend dietary habits through AI.

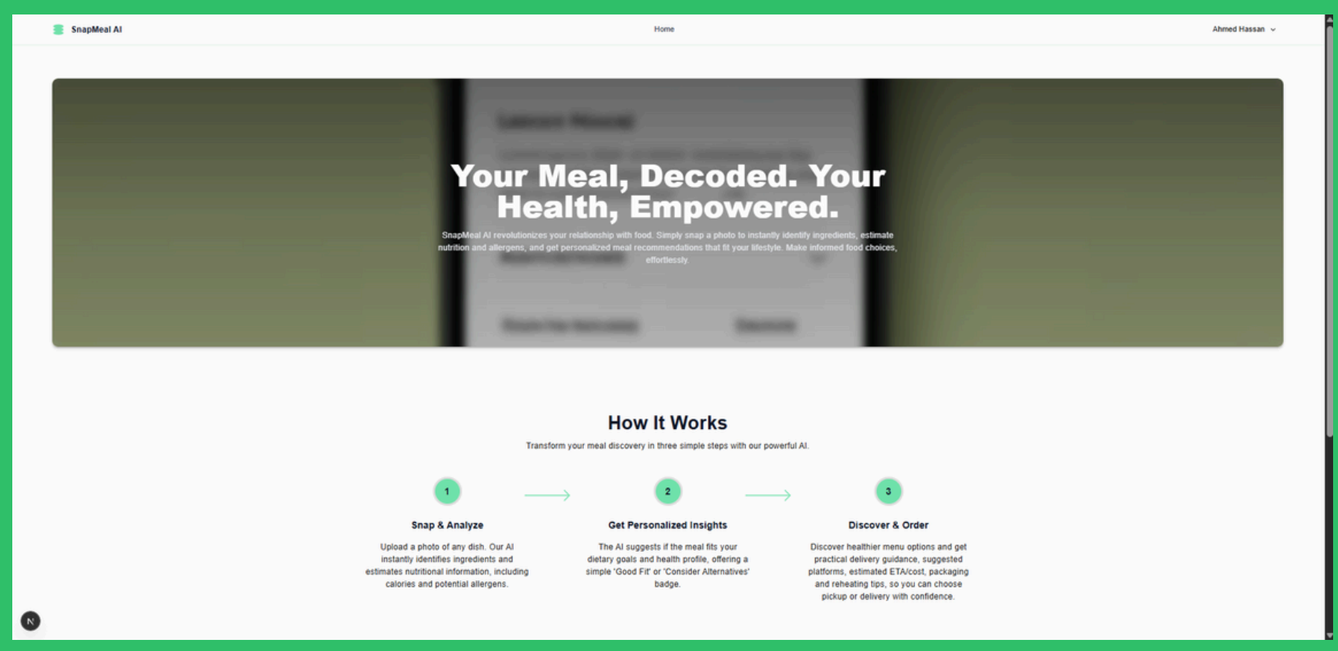
Implemented Improvements

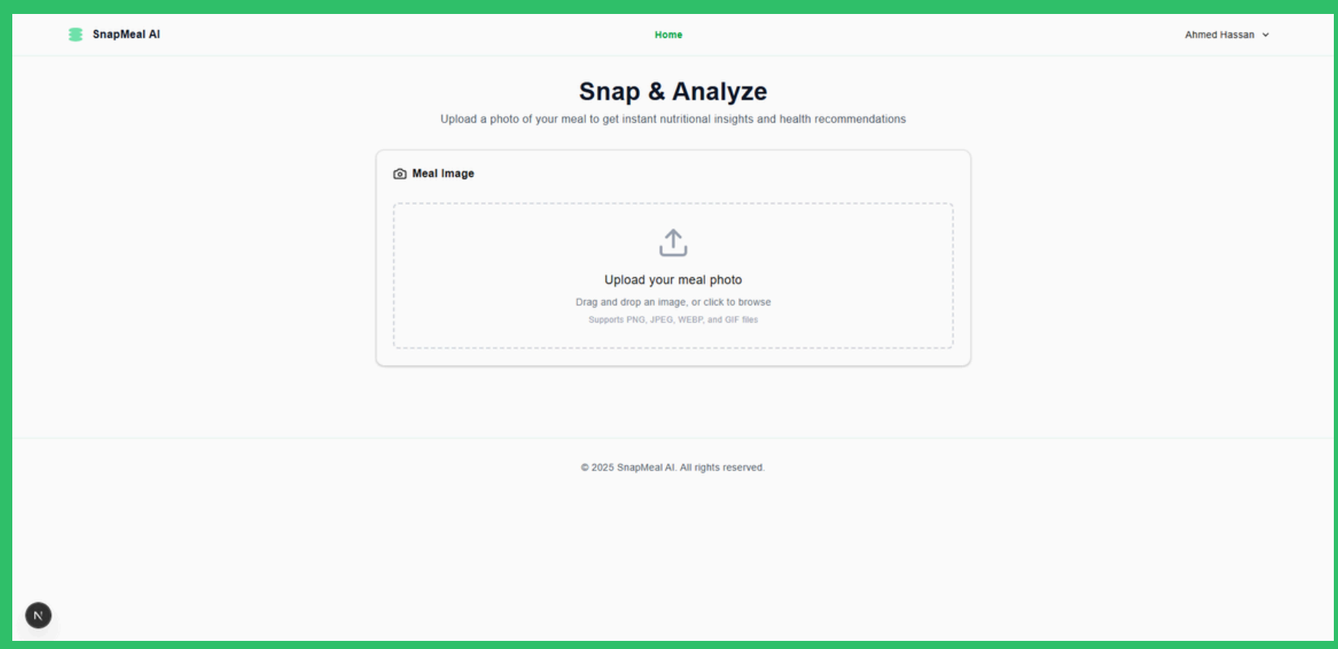
- **Authentication flow**: Signup, Login, session handling, and Home dashboard.
- **Image upload UI**: intuitive drag-and-drop or file picker for a single food-item photo (menu item or downloaded image), with client-side validation and preview.
- **GPT integration**: server-side vision + language pipeline that analyzes the uploaded image and returns structured JSON results (ingredients, nutrition summary, allergens, portion analysis, alternatives, confidence, delivery recommendations & options).
- **Results UI**: polished frontend to present nutrition summary, ingredients & details, allergen warnings, healthier alternatives, delivery recommendation, and delivery options.
- **Testing**: comprehensive quality coverage, 105 unit and integration tests with 92% code coverage to ensure reliability and robust error handling.

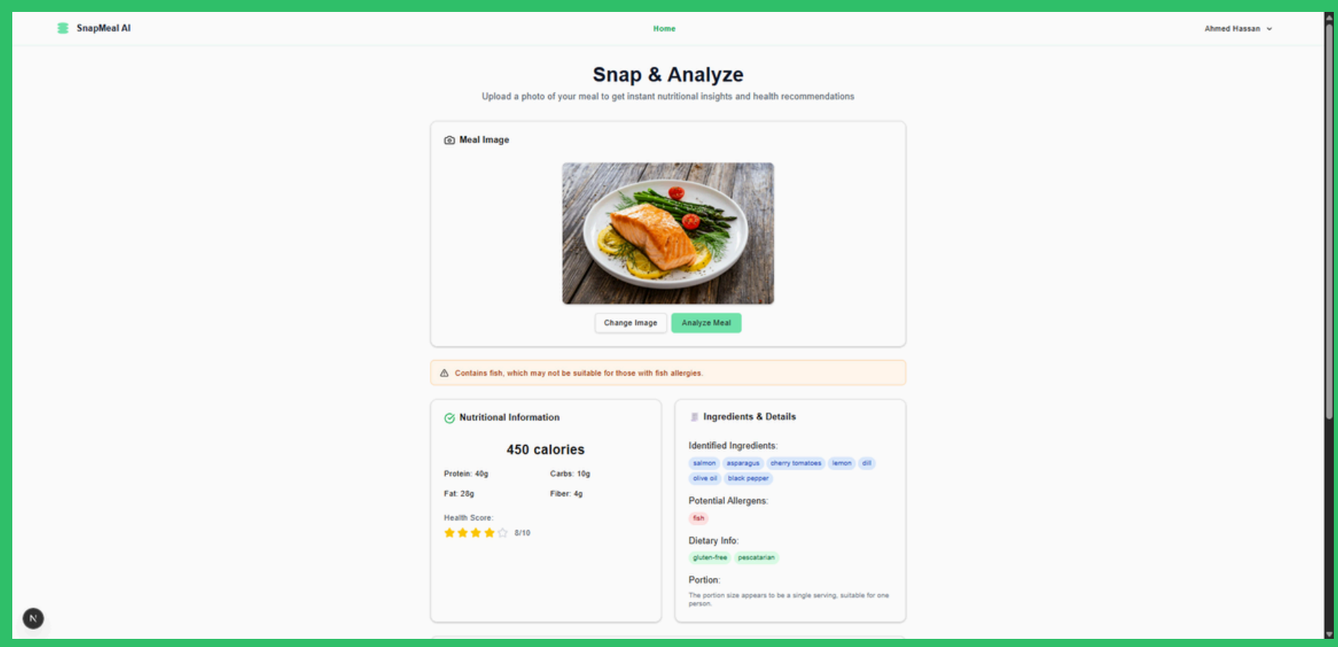
Future Implementation

Milestones

- **Camera capture**: add an in-app camera UI to capture real-time photos and submit directly for analysis.
- **History & persistence**: save past analysis results and build a searchable history page so users can review, compare, and reopen previous scans.
- **Personalized allergy preferences**: allow users to set allergy/diet preferences in their profile so the API can surface tailored allergen alerts and recommendations.
- **Admin tools**: build an admin dashboard for account management, usage monitoring, and moderation tools to support production deployment.









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